E-commerce Customer Satisfaction

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Problem Description :

in this project, we're looking at how satisfied customers are with their online shopping experiences. We're using data from Olist Stores, a big online shop in Brazil. Olist helps small businesses sell their stuff online easily. After people buy something on Olist, they get asked how happy they are with it. We're using this feedback to see if we can predict how satisfied customers will be based on their shopping data. Our goal is to find the best way to understand and predict customer satisfaction.

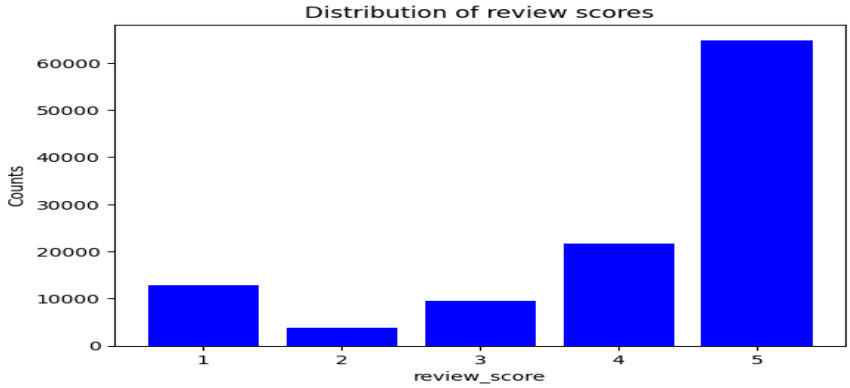
Dataset Link on kaggle

Project Pipelined :

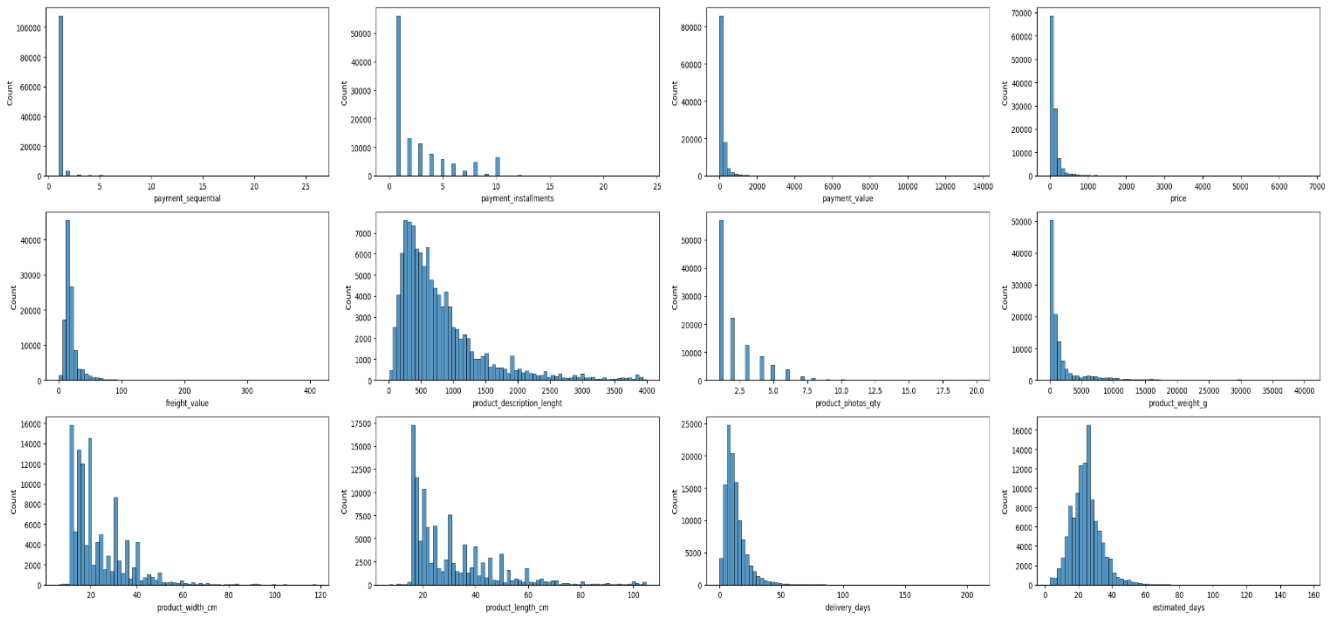
1. Data Cleaning for explore and visualize it
2. Data Exploration and get visualizations
3. Data preprocessing and get features for testing models
4. Use spark to create spark session and try different models
5. Extract Association Rules

Analysis and Solution of the Problem :

- **Data Cleaning** :
  Drop any unimportant columns, remove nulls values and mix any related columns into one columns then remove these related columns.
- **Data Exploration :**
  - ♦ Distribution of review score which shows that a large percentage of
    of customers are satisfied with services and products.

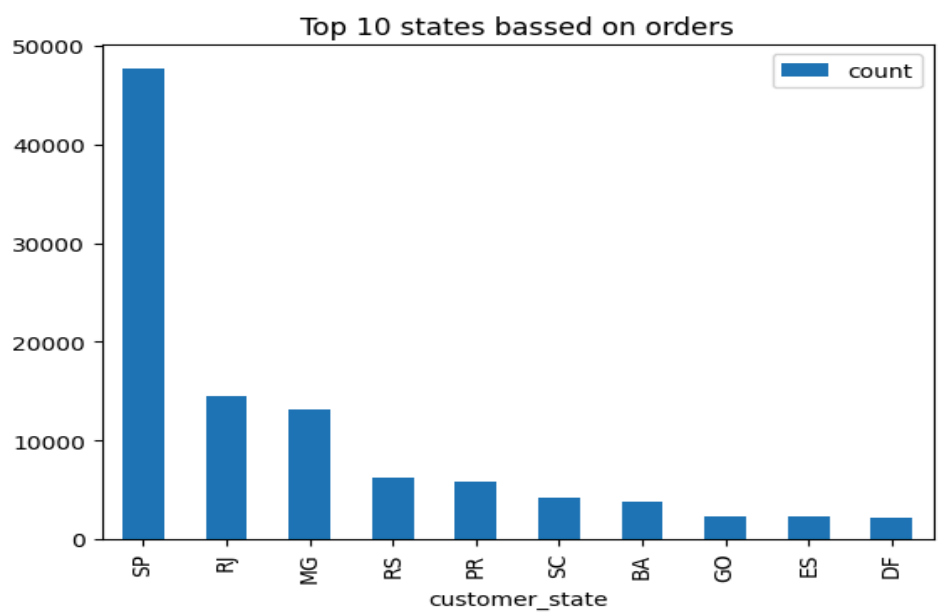


- Distribution of numerical attributes that shows what the

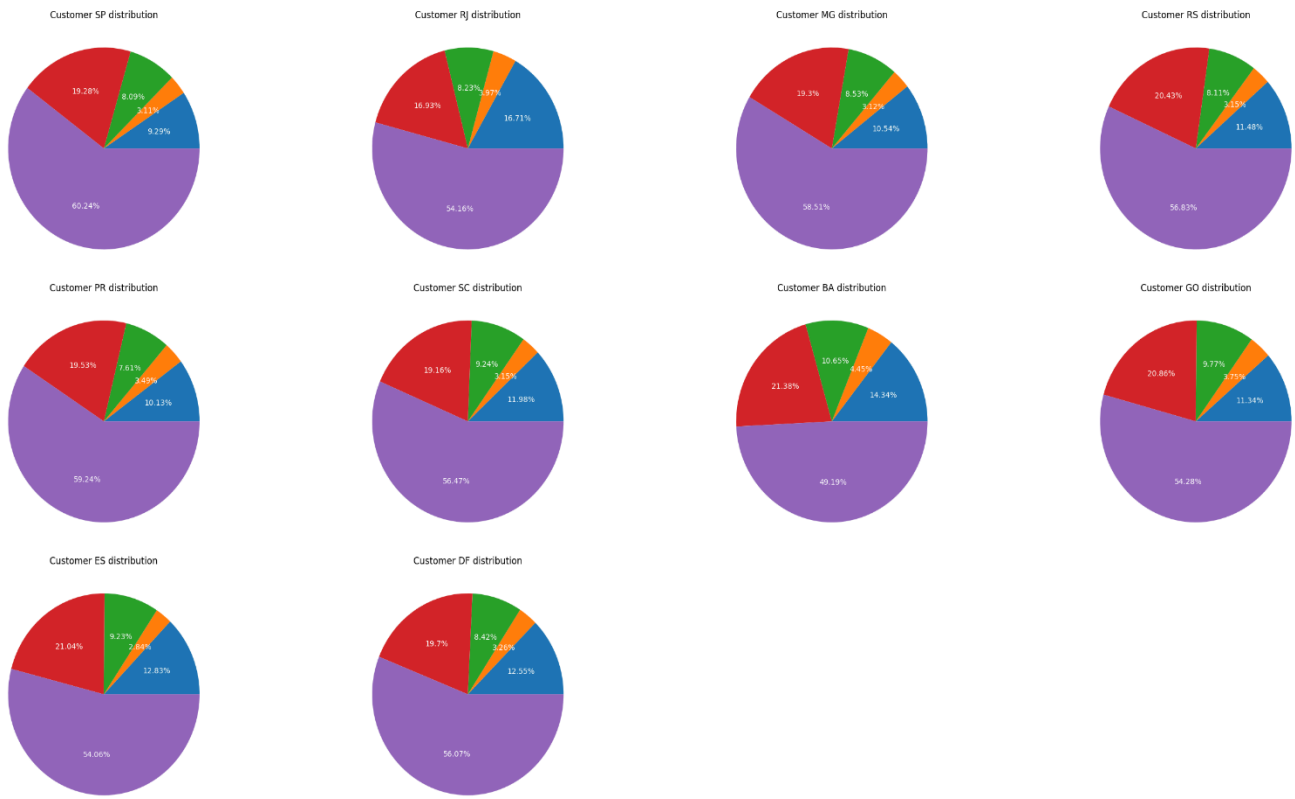


customers used more as services and also shows some distribution of returns

- Top 10 customer states



- Review scores percentage for top 10 seller states



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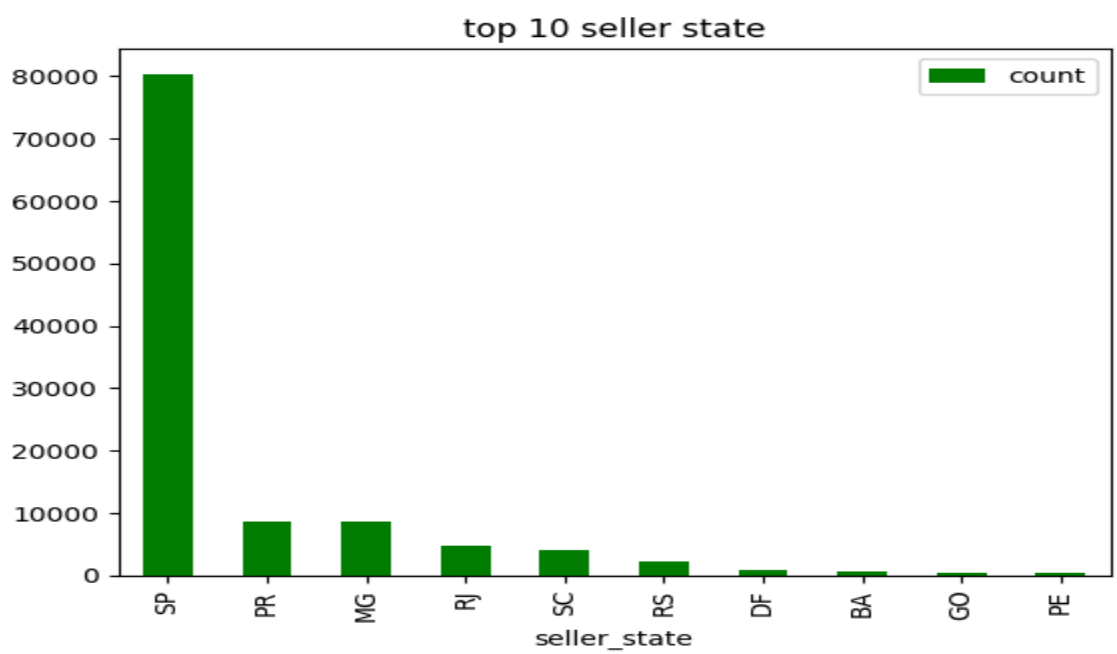

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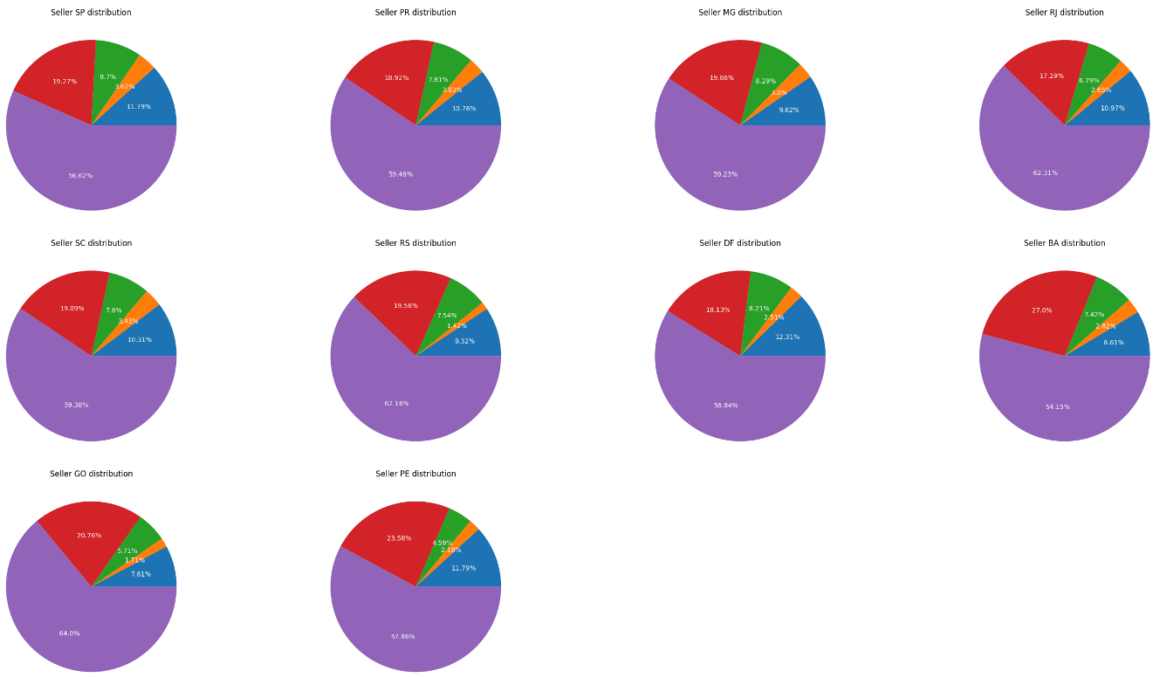
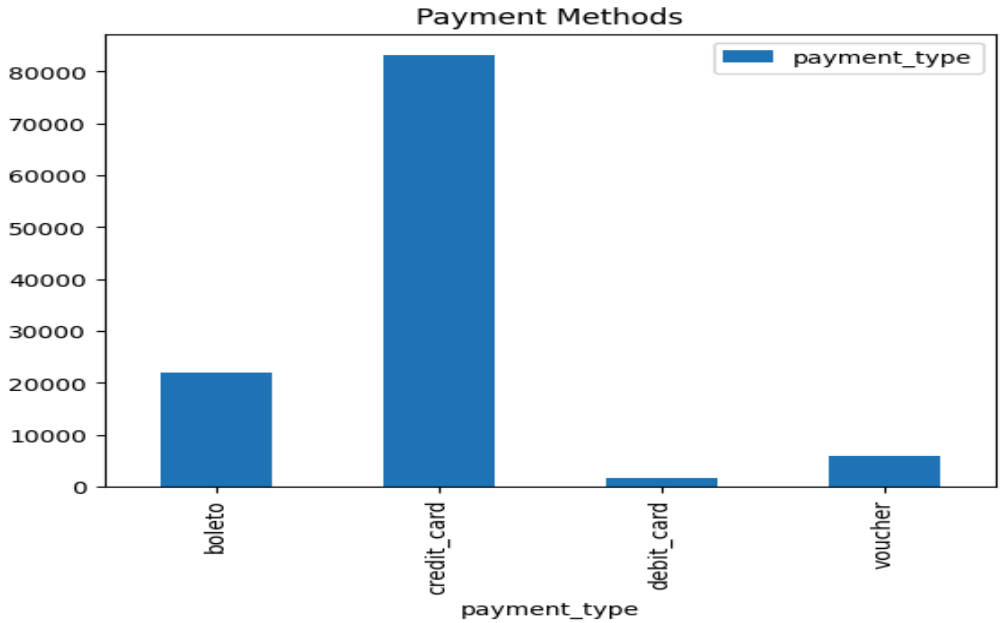
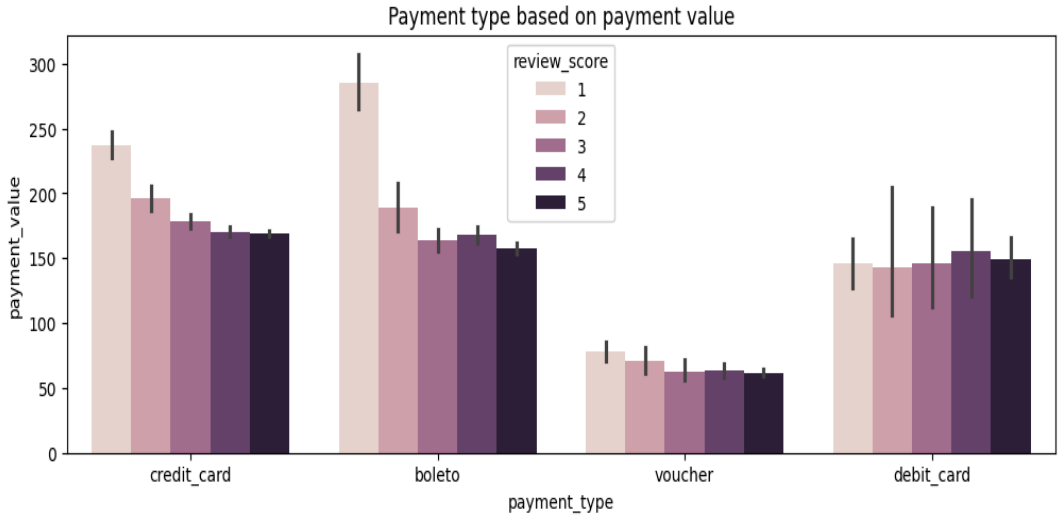


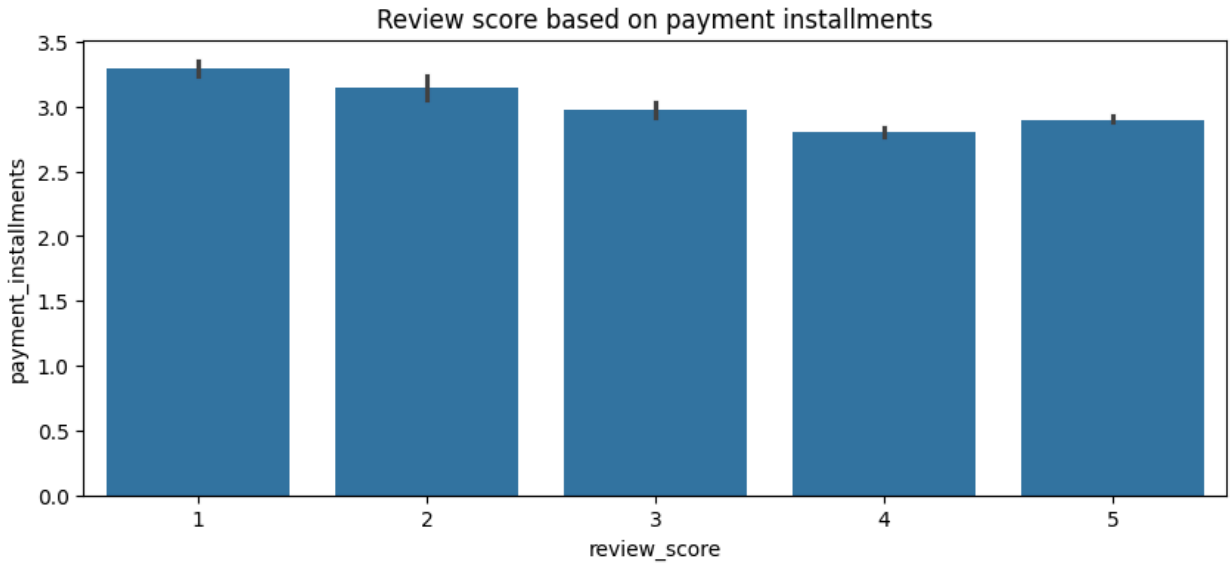
- Most used payment methods for orders



- Distribution of payment type with respect to review score

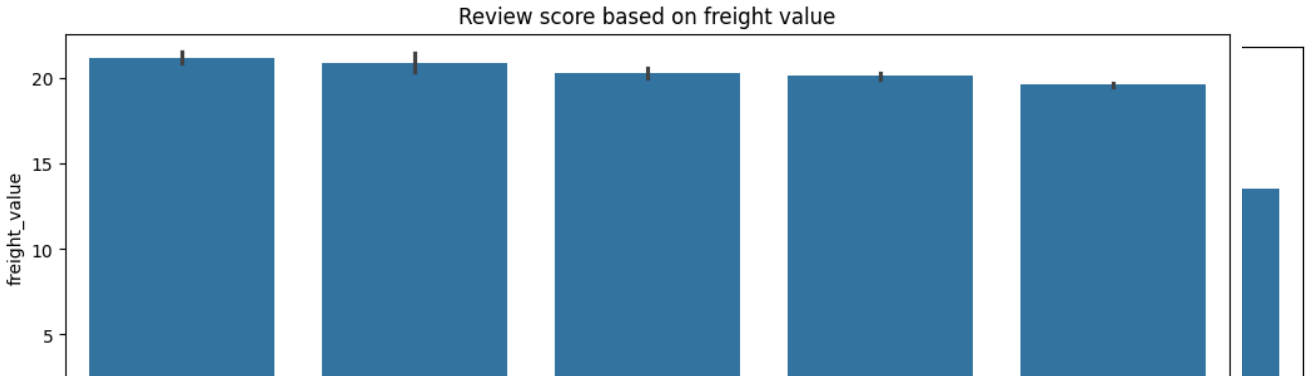
- Payment type based on payment value



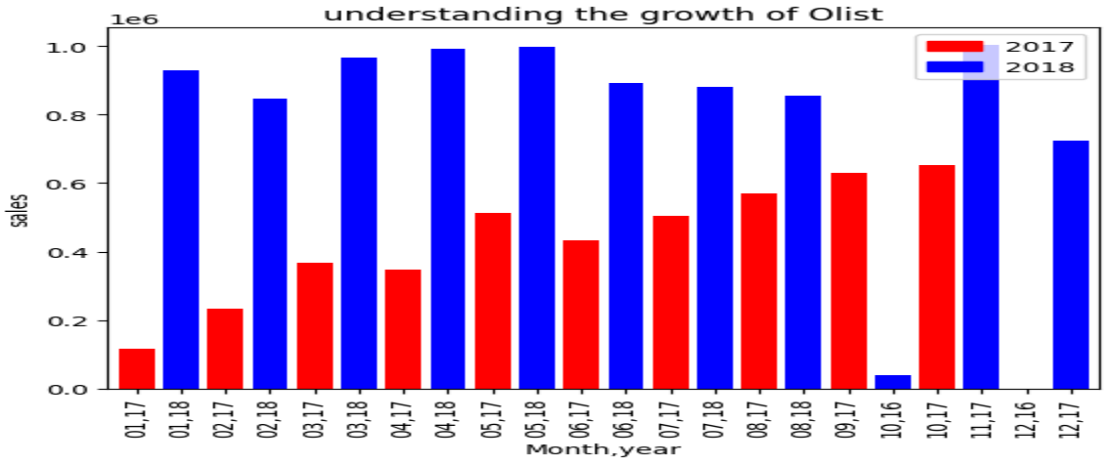


- Review score based on payment installments
- Review score based on payment value

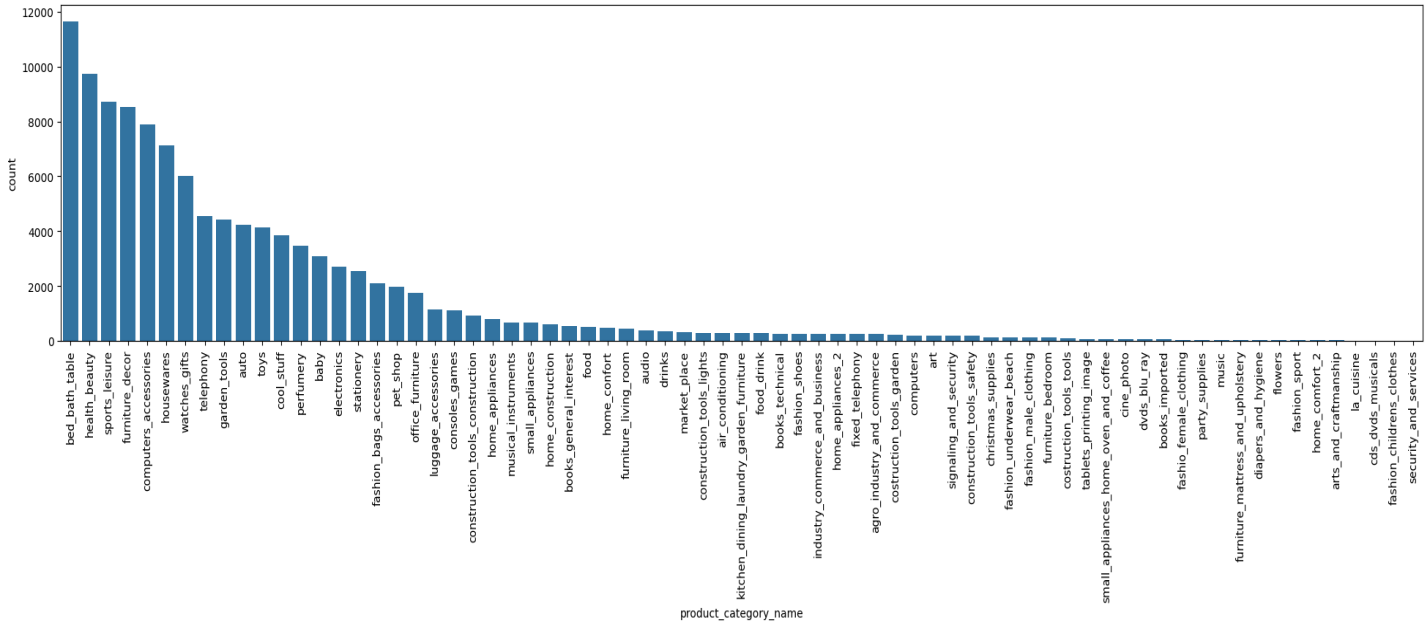
- Review score based on product price
- Review score based on freight value



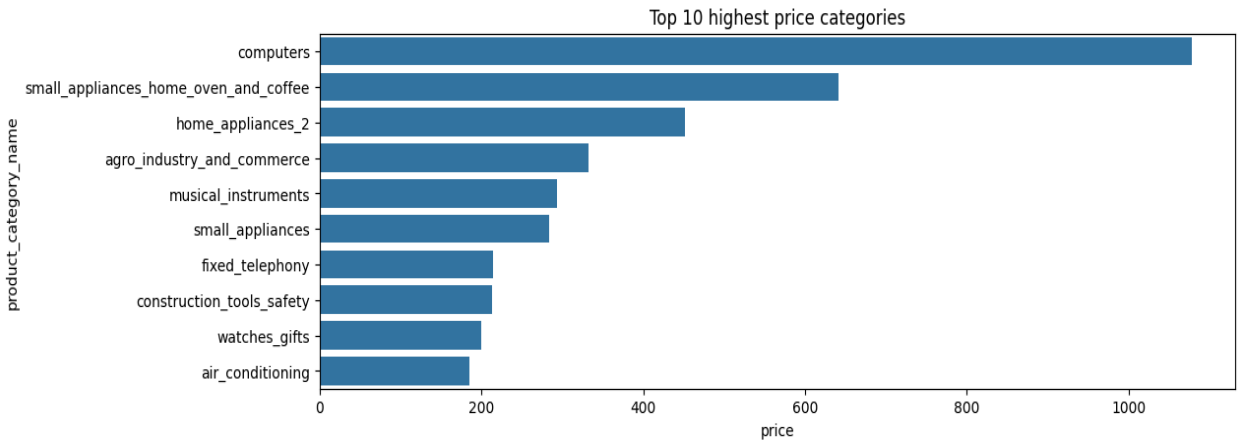
- Understand the growth of Olist



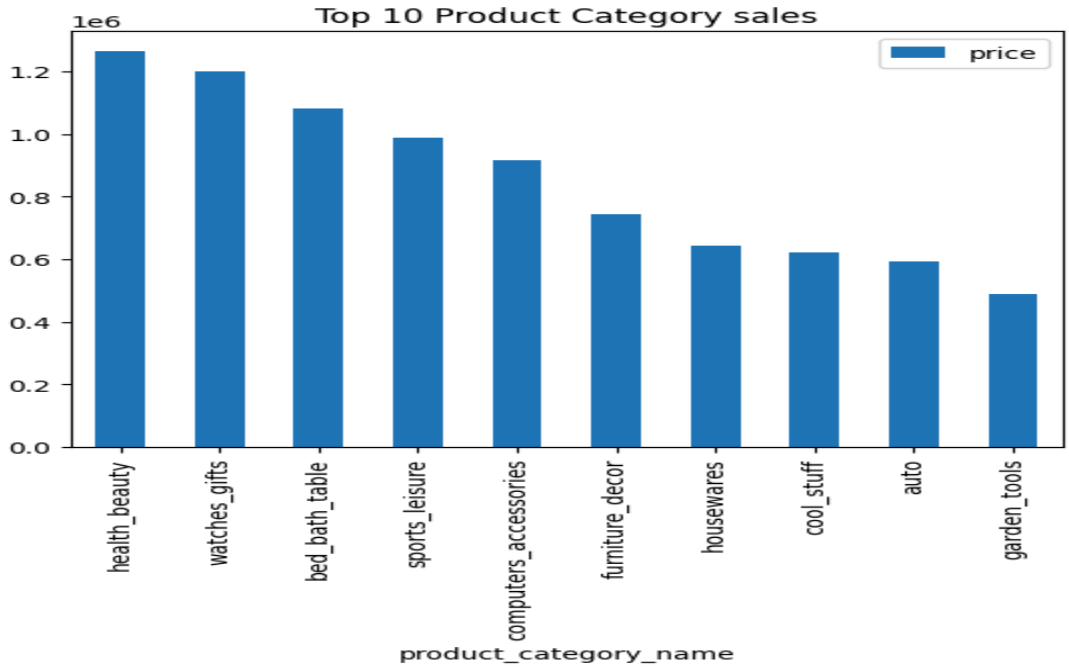
- Distribution of product category



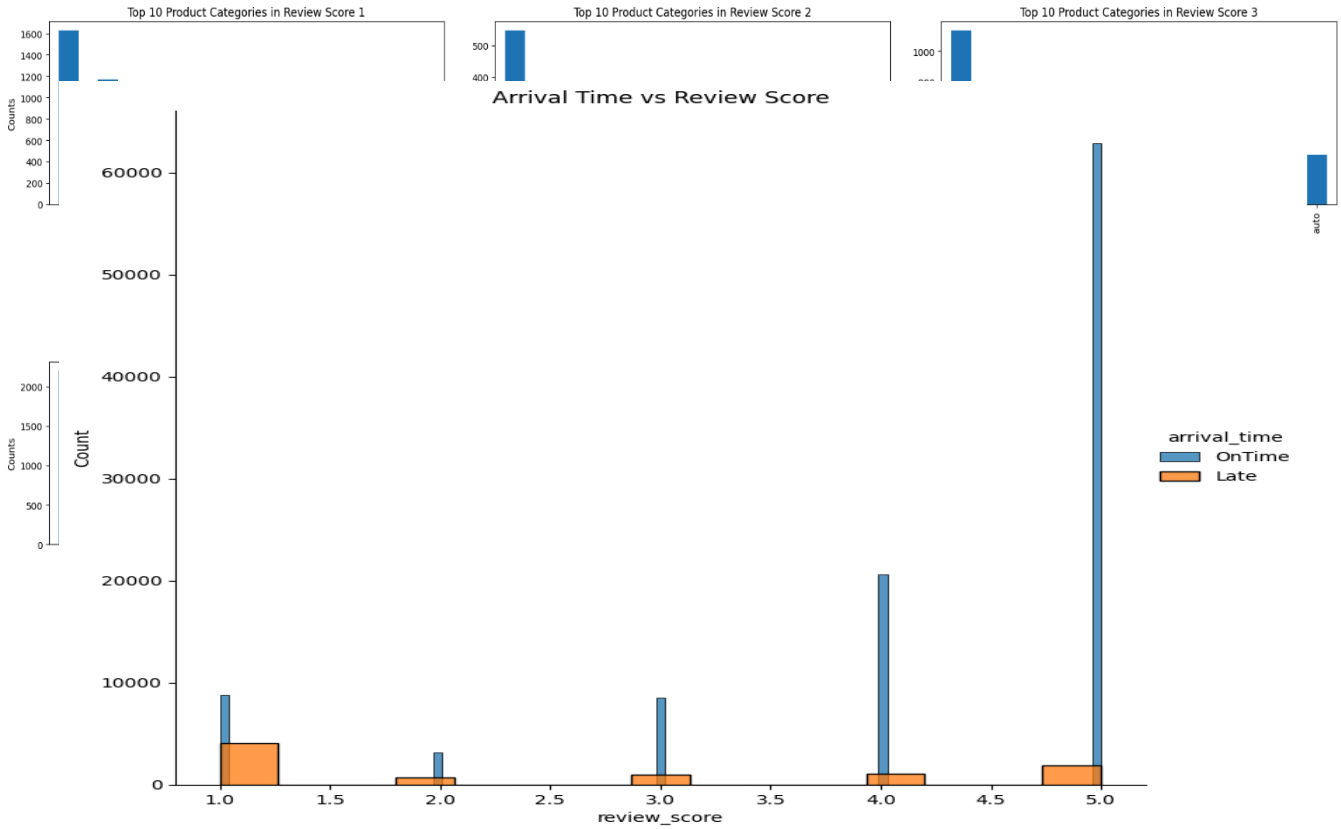
- Top 10 highest price category



- Top 10 product category sales

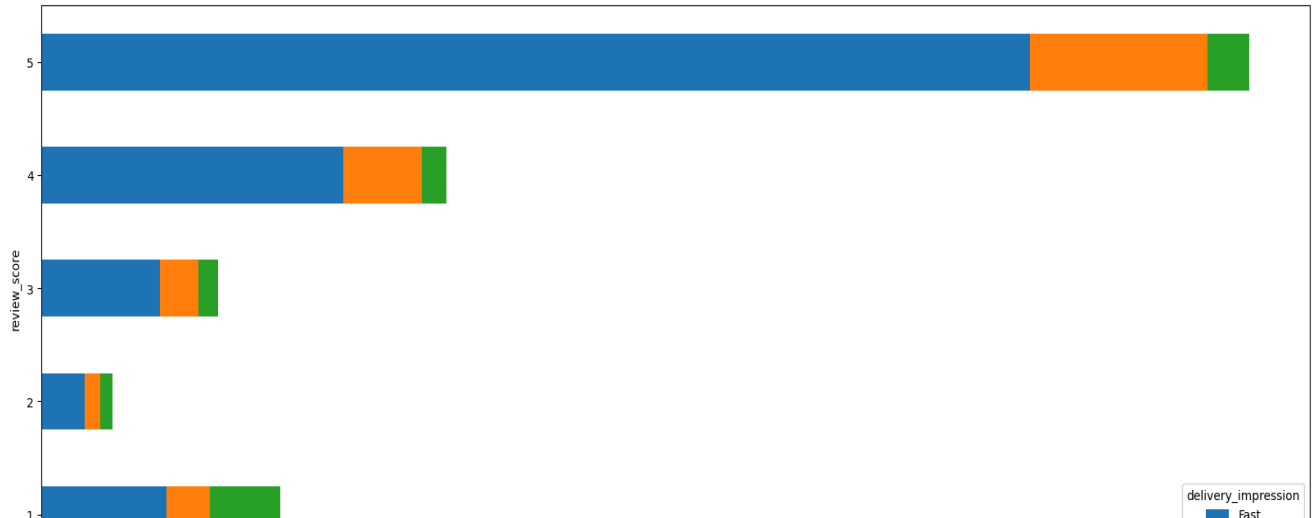


- Top 10 product categories in review score

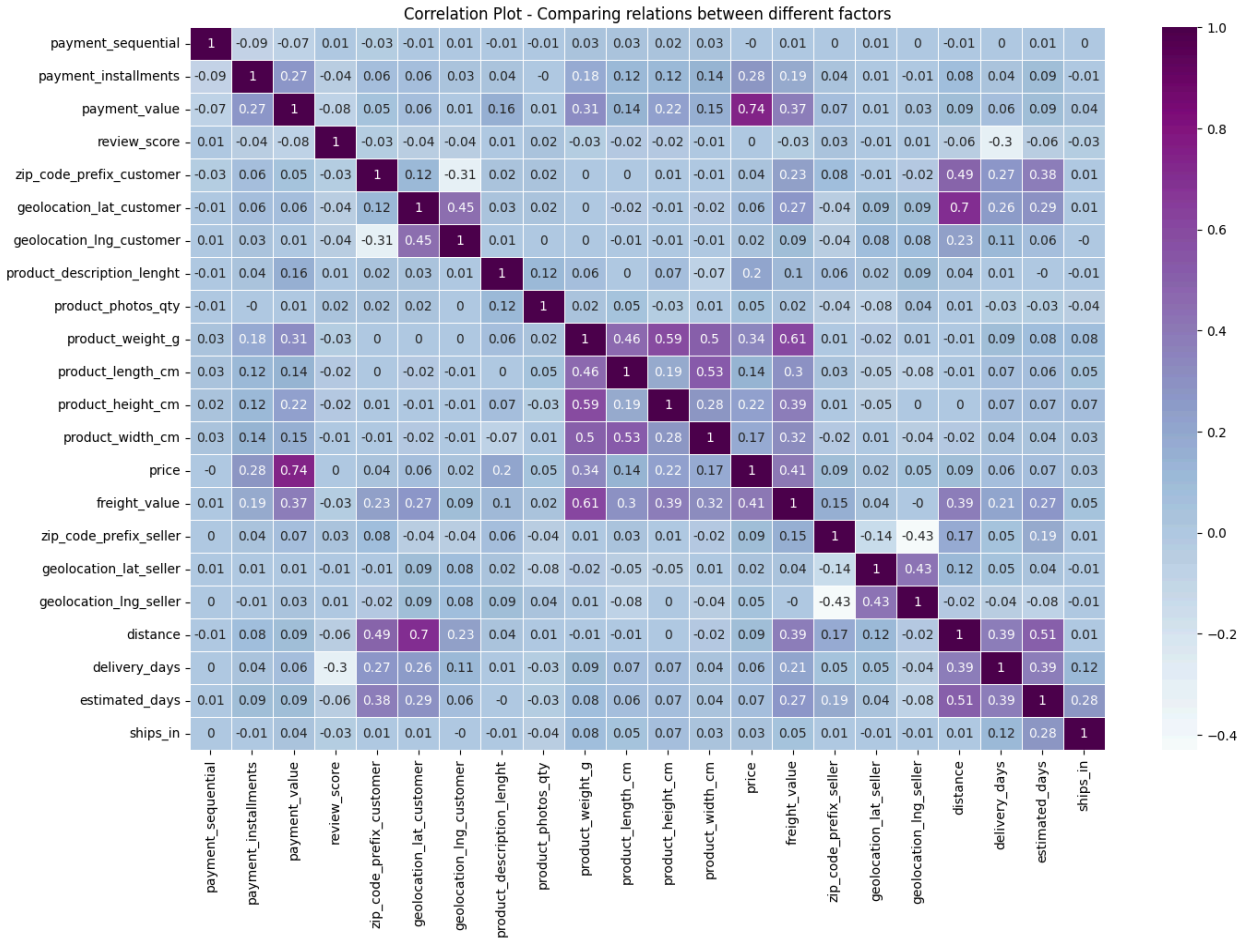


- Arrival Time vs review score

- Delivery impressions vs review score



- Correlation plot – comparing relations between different factors



- Association rules
  - items with highest support

```
sorted_by_support = frequent_itemsets.sort_values(by="support", ascending=False)
print(sorted_by_support.head(10).to_string(index=False))
```

```
 support                                                  itemsets
0.999050                                   (very small payment value)
0.998242                                             (small distance)
0.997763                                        (small freight value)
0.997292                 (very small payment value, small distance)
0.996937            (very small payment value, small freight value)
0.996129                                            (very small price)
0.996022                      (small distance, small freight value)
0.995774                 (very small payment value, very small price)
0.995197 (very small payment value, small distance, small freight value)
0.994389                                (small distance, very small price)
```

- Items with highest confidence

```
sorted_by_confidence = rules.sort_values(by="confidence", ascending=False)
print(sorted_by_confidence.head(10).to_string(index=False))
```

antecedents	consequents	antecedent support	consequent support	support	confidence	
(very small size, credit_card, OnTime, small product description lenght, very small price, small installments, small freight value, small product photos qty)	(very small payment value)	0.506579	0.99905	0.506579	1.0	1
(SP, credit_card, very small size, very small price, Fast, small freight value, small product photos qty)	(very small payment value)	0.532256	0.99905	0.532256	1.0	1
(SP, credit_card, very small size, OnTime, very small price, very small product weight, small installments)	(very small payment value)	0.471082	0.99905	0.471082	1.0	1
(SP, credit_card, very small price, small product description lenght, Fast, very small product weight, small product photos qty)	(very small payment value)	0.456388	0.99905	0.456388	1.0	1
(SP, credit_card, very small price, Fast, very small product weight, small installments)	(very small payment value)	0.491929	0.99905	0.491929	1.0	1
(small distance, SP, credit_card, very small price, small product description lenght, Fast, very small product weight, small freight value)	(very small payment value)	0.462062	0.99905	0.462062	1.0	1
(small distance, SP, very small size, credit_card, very small price, small product description lenght, small installments)	(very small payment value)	0.460215	0.99905	0.460215	1.0	1
(SP, credit_card, very small size, very small price, small product description lenght, Fast, very small product weight, small installments, small product photos qty)	(very small payment value)	0.420669	0.99905	0.420669	1.0	1
(small distance, very small price, small product description lenght, Fast, very small product weight, excellent, small product photos qty)	(very small payment value)	0.426671	0.99905	0.426671	1.0	1
(very small size, OnTime, very small price, small product description lenght, small installments, Neutral)	(very small payment value)	0.448797	0.99905	0.448797	1.0	1

- Items with highest lift

```
sorted_by_lift = rules.sort_values(by="lift", ascending=False)
print(sorted_by_lift.head(10).to_string(index=False))
```

antecedents	consequents	antecedent support	consequent support	support	confidence	lift
(small distance, very small price, very small product weight, small freight value, excellent)	(very small payment value, very small size, OnTime, Fast, small installments, small product photos qty)	0.536669	0.758697	0.453263	0.844586	1.113206
(excellent, very small product weight, small distance, very small price)	(very small payment value, very small size, OnTime, Fast, small installments, small freight value, small product photos qty)	0.536722	0.758626	0.453263	0.844502	1.113200
(excellent, very small product weight, small distance, very small price)	(very small payment value, very small size, OnTime, Fast, small installments, small product photos qty)	0.536722	0.758697	0.453281	0.844535	1.113140
(excellent, very small product weight, very small price, small freight value)	(very small payment value, small distance, very small size, OnTime, Fast, small installments, small product photos qty)	0.537521	0.757605	0.453263	0.843247	1.113043
(excellent, very small product weight, very small price)	(very small payment value, small distance, very small size, OnTime, Fast, small installments, small freight value, small product photos qty)	0.537574	0.757534	0.453263	0.843163	1.113037
(excellent, very small product weight, very small price)	(very small payment value, small distance, very small size, OnTime, Fast, small installments, small product photos qty)	0.537574	0.757605	0.453281	0.843196	1.112977
(excellent, very small product weight, very small price, small freight value)	(very small payment value, very small size, OnTime, Fast, small installments, small product photos qty)	0.537521	0.758697	0.453884	0.844403	1.112965
(excellent, very small product weight, very small price)	(very small payment value, very small size, OnTime, Fast, small installments, small freight value, small product photos qty)	0.537574	0.758626	0.453884	0.844319	1.112959
(excellent, very small product weight, very small price)	(very small payment value, very small size, OnTime, Fast, small installments, small product photos qty)	0.537574	0.758697	0.453902	0.844352	1.112898
(excellent, very small product weight, small distance, very small price)	(very small payment value, very small size, OnTime, Fast, small installments, small freight value)	0.536722	0.769759	0.459780	0.856644	1.112873

- Items with highest leverage

```
sorted_by_leverage = rules.sort_values(by="leverage", ascending=False)
print(sorted_by_leverage.head(10).to_string(index=False))
```

antecedents	consequents	antecedent support	consequent support	support	confidence	lift	leverage
(Fast, OnTime, very small payment value, very small product weight)	(small distance, very small size, very small price, small freight value, excellent)	0.802548	0.548211	0.486886	0.606675	1.106646	0.046921
(small distance, very small size, very small price, small freight value, excellent)	(Fast, OnTime, very small payment value, very small product weight)	0.548211	0.802548	0.486886	0.888137	1.106646	0.046921
(excellent, very small size, small distance, very small price)	(very small payment value, OnTime, Fast, very small product weight, small freight value)	0.548282	0.802451	0.486886	0.888022	1.106637	0.046917
(very small payment value, OnTime, Fast, very small product weight, small freight value)	(excellent, very small size, small distance, very small price)	0.802451	0.548282	0.486886	0.606749	1.106637	0.046917
(excellent, very small size, very small price, small freight value)	(very small payment value, small distance, OnTime, Fast, very small product weight)	0.549072	0.801323	0.486886	0.886744	1.106600	0.046902
(very small payment value, small distance, OnTime, Fast, very small product weight)	(excellent, very small size, very small price, small freight value)	0.801323	0.549072	0.486886	0.607603	1.106600	0.046902
(very small payment value, small distance, OnTime, Fast, very small product weight, small freight value)	(excellent, very small size, very small price)	0.801234	0.549152	0.486886	0.607670	1.106561	0.046887
(excellent, very small size, very small price)	(very small payment value, small distance, OnTime, Fast, very small product weight, small freight value)	0.549152	0.801234	0.486886	0.886615	1.106561	0.046887
(excellent, very small size, small distance, very small price)	(Fast, OnTime, very small payment value, very small product weight)	0.548282	0.802548	0.486904	0.888054	1.106543	0.046881
(Fast, OnTime, very small payment value, very small product weight)	(excellent, very small size, small distance, very small price)	0.802548	0.548282	0.486904	0.606698	1.106543	0.046881

- Payment type and review score

```
excellent -> credit_card
credit_card -> excellent
```

- Payment value and review score
  excellent -> very small payment value
  very small payment value -> excellent

- customer state and review score

  SP -> excellent

- seller state and review score
  excellent -> SP
  SP -> excellent

- Arrival time and review score
  excellent -> OnTime
  OnTime -> excellent

- Delivery impression and review score

  Fast -> excellent
  excellent -> Fast

- **Models / Results and Evaluation**

Model	Accuracy	F1_Score
Random Forrest	59.9	47.3
Logistic regression	59.7	47.4
Decision Tree	60	47.3
Naïve Bayes	54	47

- **Unsuccessful trials**
  - SVM model takes long time in training
  - When I reduce features using PCA, it gives me low accuracy

- **Future Work**
  - We will work on features and try to increase accuracy